The Banach Fixed-Point Theorem: From a Calculator Trick to Spaces of Functions

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1 A calculator puzzle

Take any calculator, type in a number, and repeatedly press the cosine button, working in radians. Start, say, with $x_0 = 1$, and compute $x_1 = \cos(x_0)$, then $x_2 = \cos(x_1)$, and so on. No matter what number you start with, the sequence converges to the same value,

$$x \approx 0.739085\ldots,$$

which is the unique solution of the equation

$$x = \cos(x).$$

Such a number is called a *fixed point* of the function $f(x) = \cos(x)$: feeding it into f returns the same number. The puzzle is why this always happens, regardless of the starting point, and what property of the cosine function is responsible. The answer is the Banach fixed-point theorem, and building it up from ordinary single-variable calculus is the purpose of this essay. Along the way the same argument will be shown to generalize far beyond the real line, to the point where it underwrites the existence and uniqueness of solutions to differential equations.

2 Contractions and the Mean Value Theorem

Suppose f is differentiable and its derivative satisfies $|f'(x)| \leq k$ for some constant $k < 1$ on some interval. The Mean Value Theorem says that for any two points x, y in that interval there is a point c between them with

$$f(x) - f(y) = f'(c)(x - y),$$

and hence

$$|f(x) - f(y)| \leq k\,|x - y|.$$

In words: f shrinks the distance between any two points by a factor of at least k. A function with this property is called a *contraction.*

The cosine function is a contraction near its fixed point. Its derivative is $f'(x) = -\sin(x)$, so $|f'(x)| \leq 1$ everywhere, and in fact $|f'(x)| < 1$ away from the origin. Near the fixed point $0.739\ldots$, one has $\sin(x) \approx 0.67$, comfortably below 1. This is the entire mechanism behind the calculator experiment: cosine contracts distances, and contractions force convergence, as the next section shows.

3 Why a contraction forces convergence

Define the iteration $x_{n+1} = f(x_n)$ starting from an arbitrary point x_0, where f is a contraction with constant $k < 1$. Look at the distance between consecutive terms:

$$|x_2 - x_1| = |f(x_1) - f(x_0)| \leq k\,|x_1 - x_0|,$$

$$|x_3 - x_2| = |f(x_2) - f(x_1)| \leq k\,|x_2 - x_1| \leq k^2\,|x_1 - x_0|,$$

and in general

$$|x_{n+1} - x_n| \leq k^n\,|x_1 - x_0|.$$

The steps between successive terms shrink geometrically. This is the familiar situation from geometric series: if the step sizes decay like k^n with $k < 1$, then the total distance still to be travelled from step n onward is bounded by a convergent geometric series,

$$|x_{n+m} - x_n| \leq \sum_{i=n}^{n+m-1} |x_{i+1} - x_i| \leq k^n\,|x_1 - x_0| \sum_{i=0}^{\infty} k^i = \frac{k^n}{1-k}\,|x_1 - x_0|,$$

and the right-hand side tends to 0 as $n \to \infty$. The tail of the sequence is therefore squeezed into an arbitrarily small interval: the sequence has nowhere left to wander, so it must converge to some limit L. Since f is continuous, taking the limit of both sides of $x_{n+1} = f(x_n)$ gives $L = f(L)$, so the limit is itself a fixed point.

Uniqueness follows from the same contraction inequality. If L_1 and L_2 were both fixed points, then

$$|L_1 - L_2| = |f(L_1) - f(L_2)| \leq k\,|L_1 - L_2|,$$

which, since $k < 1$, is only possible if $|L_1 - L_2| = 0$. So a contraction has at most one fixed point, and the iteration finds it starting from anywhere. The geometric-series bound above is not just a convergence proof; it is also an explicit error estimate, giving the number of iterations needed to approximate the fixed point to any desired tolerance without knowing the fixed point in advance.

4 Generalizing beyond the real line

Nothing in the argument of the previous section used the fact that the x_n were real numbers. The only ingredients were:

- a notion of distance between two points, written $|x - y|$ on the real line, satisfying the usual triangle inequality;
- the fact that a sequence whose terms get arbitrarily close together must converge to a point that actually belongs to the space in question, a property known as *completeness* (the real numbers have it; the rational numbers famously do not, since the decimal approximations to $\sqrt{2}$ get arbitrarily close together without converging to a rational number);
- a map f that shrinks distances by a fixed factor $k < 1$.

None of this requires the points to be numbers. They could equally well be points in $\mathbb{R}^n$, with distance measured by the Euclidean norm and the contraction condition checked through the Jacobian rather than a single derivative. They could even be entire *functions*. As long as the collection of objects in question comes equipped with a sensible notion of distance, and is complete with respect to that distance, the same telescoping-series argument goes through unchanged. This is the content of the general theorem.

Theorem (Banach fixed-point theorem). Let (X, d) be a complete metric space and let $f : X \to X$ satisfy

$$d(f(x), f(y)) \leq k\, d(x, y)$$

for some fixed constant $k < 1$ and all $x, y \in X$. Then f has exactly one fixed point in X, and for any starting point $x_0 \in X$, the sequence defined by $x_{n+1} = f(x_n)$ converges to that fixed point.

The abstraction is not idle. It is exactly what is needed to make sense of iteration on objects that are not numbers at all, and the most important example of such an object is a function.

5 The space of functions

Consider two continuous functions $\varphi(x)$ and $\psi(x)$ defined on an interval $[a, b]$. Their distance can be defined as the largest vertical gap between their graphs,

$$d(\varphi, \psi) = \max_{x \in [a,b]} |\varphi(x) - \psi(x)|.$$

This is a perfectly legitimate notion of distance: two functions are close if their graphs stay close together everywhere on the interval, and far apart if their graphs diverge substantially anywhere, even at a single point. Equipped with this distance, the collection of continuous functions on $[a, b]$ becomes a complete metric space in exactly the sense required by the theorem: it has a notion of distance, and a sequence of functions that get uniformly closer and closer together does converge to another continuous function on $[a, b]$. This last fact plays the same role that completeness of the real numbers plays in ordinary calculus; it can be taken on faith at this stage, in the same way that the Intermediate Value Theorem is used without re-deriving the completeness of $\mathbb{R}$ each time.

With this notion of distance in hand, a map that takes a function as input and returns another function as output can be a contraction in exactly the same sense as before, and the fixed point of such a map is not a number but a function.

6 Application: existence and uniqueness of solutions to ordinary differential equations

This machinery is precisely what underlies the Picard–Lindelöf theorem. Consider an initial value problem

$$y' = f(x, y), \qquad y(x_0) = y_0.$$

By the Fundamental Theorem of Calculus, this is equivalent to the integral equation

$$y(x) = y_0 + \int_{x_0}^{x} f(t, y(t))\, dt.$$

Define an operator T that takes a candidate function φ and returns a new function $T\varphi$, given by

$$(T\varphi)(x) = y_0 + \int_{x_0}^{x} f(t, \varphi(t))\, dt.$$

A solution of the original differential equation is precisely a function y satisfying $Ty = y$: a fixed point of T, in the sense of the previous section, except that the "point" being sought is an entire curve rather than a single number.

Suppose $f(x, y)$ satisfies a Lipschitz condition in its second argument, meaning there is a constant L such that $|f(x, \varphi) - f(x, \psi)| \le L\,|\varphi - \psi|$ for all relevant φ, ψ. Then for two candidate functions φ and ψ,

$$|(T\varphi)(x) - (T\psi)(x)| = \left| \int_{x_0}^{x} \big[f(t, \varphi(t)) - f(t, \psi(t))\big]\,dt \right| \le \int_{x_0}^{x} L\,|\varphi(t) - \psi(t)|\,dt.$$

Since $|\varphi(t) - \psi(t)| \le d(\varphi, \psi)$ for every t in the interval, by definition of the sup-norm distance, the integral on the right is bounded by $L\,|x - x_0|\,d(\varphi, \psi)$. Restricting attention to a sufficiently small interval of width h around x_0, chosen so that $Lh < 1$, gives

$$d(T\varphi, T\psi) \le \underbrace{Lh}_{k<1}\, d(\varphi, \psi).$$

This is exactly the contraction inequality from Section 2, transported from the real line to the space of continuous functions. Everything that followed from it there follows here as well. Iterating $\varphi_{n+1} = T\varphi_n$ from any starting function, typically the constant function $\varphi_0(x) = y_0$, produces a sequence of functions whose distances shrink geometrically; by the argument of Section 3, this sequence converges to a function y satisfying $Ty = y$, which is the unique solution of the initial value problem on the chosen interval. The successive iterates $\varphi_0, \varphi_1, \varphi_2, \ldots$ are precisely the Picard iterates, and the geometric-series bound gives an explicit estimate of how closely a given iterate approximates the true solution.

7 Concluding remarks

The path from a calculator experiment with cosine to the existence and uniqueness of solutions of differential equations passes through exactly one idea, applied twice. A contraction shrinks distances by a fixed factor less than one; iterating a contraction produces a sequence whose steps shrink geometrically; and a geometric sequence of shrinking steps has nowhere to go but toward a single point, which must be the unique fixed point of the map. On the real line this point is a number, found by an idle sequence of button presses. On the space of continuous functions, equipped with the natural notion of distance between two graphs, the same argument produces a function, and that function is the solution to a differential equation. The abstraction from numbers to functions costs nothing in the proof and gains a great deal in reach: the same page of reasoning that explains why repeated cosine always lands on $0.739\ldots$ also explains why a well-posed initial value problem has exactly one solution.